

- Please write your answers legibly and neatly.
- Symbols, constants, terminology used have their usual meaning unless specified specially.
- There are a total of **SIX** questions and all are compulsory. Each carries marks as indicated
- Answer the **sub-parts** of a question at one place and in the order in which they appear.
- Preferably start each question on a **fresh page** however sub parts can be written in continuation.

For 0.25 μ m Technology node --
Take $V_{dd}=2.5$ V, $V_{TON}=0.4$ V, $V_{TOP}=-0.43$ V, $L_{min}=250$ nm, $C_{OX}=6$ fF/ μ m², $\lambda_n=0.05$ V⁻¹,
 $\lambda_p=0.05$ V⁻¹, $\mu_n C_{OX}=115$ uA/V², $\mu_p C_{OX}=30$ uA/V²

For digital gates--- $R_{nmos} = 13 \text{ kohm}/(W/L)$, $R_{pmos} = 31 \text{ kohm}/(W/L)$,
For short channel MOSFET (additional parameters)--- $V_{sat} = 10^5 \text{ m/sec}$, $V_{DSATn} = 0.63 \text{ V}$, $V_{DSATp} = -1 \text{ V}$,

- Use long channel approximation and lambda based design rules unless specially mentioned.
- Use square law current equation unless specifically mentioned.
- Neglect body effect, channel length modulation if not mentioned specifically.
- The body of all PMOS is tied to V_{DD} while all NMOS is tied to ground unless specially mentioned.

Q 1. For bandgap reference circuitry shown in Fig 1a, assume all the resistances are independent of temperature. The aspect ratio of transistor M_1 , M_2 and M_3 are equal, and emitter area of bipolar transistor Q_1 is 20 times emitter area of transistor Q_2 ($A_{E1}:A_{E2} = 20:1$).

a) Find out the ratio $R_A:R_1$ for zero temperature coefficient (ZTC) of reference voltage (V_{REF}).

b) If $R_1 = R_3 = 1 \text{ k}\Omega$, what will be the value of V_{REF} for ZTC.

c) If $R_1 = R_3 = 1\text{ k}\Omega$ and $R_A = 4\text{ k}\Omega$, find out temperature coefficient of V_{REF} in $\text{PPM}/^\circ\text{C}$.

d) Fig. 1b shows reference voltage generation using two diode connected MOSFET. Assuming $\mu_n C_{ox}(W/L)_n = \mu_p C_{ox}(W/L)_p$ and $V_{Tn} = |V_{Tp}|$. Find out the reference voltage V_{REF} and sensitivity of reference voltage to supply voltage.



- Q 2. For the Operational amplifier (op-amp) circuit shown in Fig 2. Assume all transistors have $V_{GS} = V_T = 0.2V$, and $C_X = 10 \text{ fF}$, $C_{\text{vout1}} = 0.1 \text{ pF}$, $C_L = 0.1 \text{ nF}$, $I_1 = I_9 = I_{13}$.

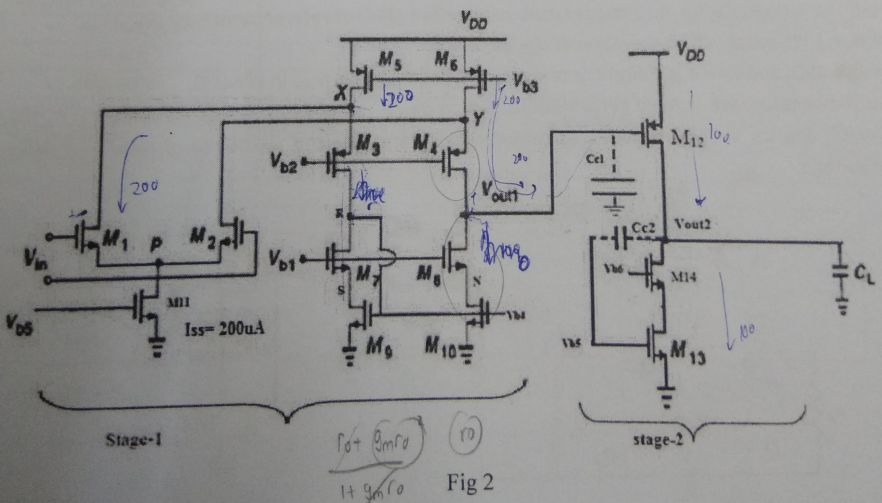
For (op-amp) in open loop mode. (Without C_{c1} , and C_{c2}) --

- Determine the value of g_m and r_o of each transistor
- Intuitively, write the expression and determine the value of the following low frequency small signal gain $[V_{\text{out1}}/V_{\text{in}}]$, and $(V_{\text{out2}}/V_{\text{out1}})$. Find total gain $[= v_{\text{out2}}/v_{\text{in}}]$, round it down to nearest multiple of 10.
- Intuitively, write the expression and then determine the value of pole frequency at the output of each stage. Hence sketch & label corner Bode magnitude plot. Mark -3dB and unity gain frequency ' ω_o ' on it.
- Determine response time of opamp in normal mode operation of both stages. Take that total response time, with stabilization, required for final output is 5 τ .
- Determine slew rate of stage-1 and stage-2. Assume full voltage swing [0 to 2.5V] at v_{out1} , v_{out2} nodes.
- Now determine the value of the total response time of opamp in slew mode operation of both stages.

Now op-amp of Fig. 2 is configured in negative feedback mode with feedback factor =1,

- Using results of part (a), part (b) and part (c), determine phase margin without C_{c1} , and C_{c2} .
- Capacitors C_{c1} , and C_{c2} are used to stabilize the op-amp. If $C_{c1} = 0.9 \text{ pF}$, approximately what value of C_{c2} is required to obtain a phase margin of 45° ?

[20]



Q3. Implement the given logic function $F = AB + CD$

$$F = AB + CD$$

- Using static CMOS topology. (Note: Complementary inputs and inverters are not available)
- Now, draw an equivalent inverter of each gate in the above circuit for worst case and size individual transistors for equal worst case pull-up and pull-down delay. (Given: minimum-sized (W/L) nmos is $3\mu\text{m}/1\mu\text{m}$, $V_{tn} = |V_{tp}| = 0.4\text{V}$, and $\mu_n C_{ox} = 90\text{ uA/V}^2$, $\mu_p C_{ox} = 30\text{ uA/V}^2$)
- Hence, size the individual transistors of each gate such that the worst case pull-up and pull down delays are equal
- Now, calculate propagation delay (from input to out using average current method considering $C_L = 150\text{ fF}$ output capacitive load. Also, calculate maximum dynamic power dissipation.
- Determine the logical effort for all the inputs.

Q4. Consider the circuit in Fig. 4, which consists of an NMOS device and PMOS current source load. (dimensions are in μm)

Given parameters are $V_{tn} = |V_{tp}| = 0.5\text{ V}$, $\mu_n = 2\mu\text{p} = 220\text{ cm}^2/\text{V-s}$, $C_{ox} = 0.23\text{ fF}/\mu\text{m}^2$ (both N & P MOS), $\lambda_n = \lambda_p = 0.1\text{ V}^{-1}$.

$$g = 23\text{ pF}/\mu\text{m}^2$$

- Calculate the width of the PMOS device considering the saturation current of $50\mu\text{A}$.
- Calculate V_M , V_{OH} , V_{OL} assuming $V_{OH} = V_{MAX}$, and $V_{OL} = V_{MIN}$.
- Calculate the low frequency small signal voltage gain $[= v_{out}/v_{in}]$ of this circuit in terms of trans-conductance and output resistance, when $V_{in} = V_M$.

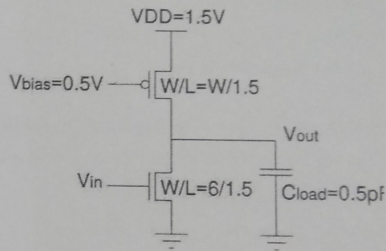


Fig 4

Q5. For the circuit shown in Fig. 5, --

CL-n ($n = 1, 2, 3, 4$) represents the combinational logic blocks and F-n ($n = 1, 2, 3, 4$) represents the memory [FLIP FLOP] elements.

The values of skews are in the ratio $\text{tskew1} : \text{tskew2} : \text{tskew3} = 1:1:2$.

For memory element, Given set-up time = 50 ps , hold time = 200 ps and

t_{c-q} (propagation delay for the input to appear at output in memory element) = 300 ps

- If the minimum clock cycle for the path CL-1 - CL-2 is 440 ps , find the values of tskew1 , tskew2 and tskew3 . $F1$ $F2$
Verify if [CL-1 - CL-2] path satisfies the hold time constraint. $F2$ $F3$

- If the minimum clock cycle for the [CL-2 - CL-3] path is $6 * \text{tskew3}$, find the value of x , x being the maximum delay in CL-2.

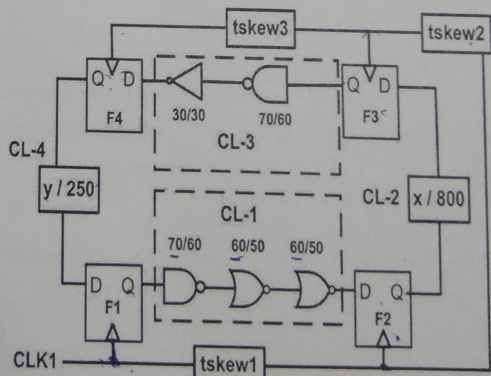


Fig 5

- c) Verify the hold time constraint for the path [CL-4 – CL-1]. If it is not satisfied, apply appropriate but minimal modification. For the modified path, find the minimum clock cycle for path CL-3 – CL-4 if $y = x$.

$F1 - F3$

Note---All timing values in the figure are in picoseconds [i.e 70/60 or x/80 implies $t_{max}(ps)/t_{min}(ps)$]

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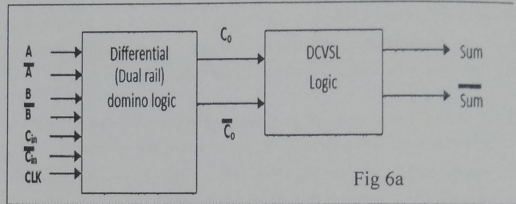
Q 6. For Fig. 6a

- a) One bit- full adder is represented by the equations (given below)..

$$Sum = A \oplus B \oplus C_{in} + \overline{C_o} (A+B+C_{in})$$

$$C_o = AB + C_{in} (A+B)$$

Realize the schematic of the full adder as shown in the block diagram (Fig 6a)



- b) Now a register (flip flop) based pipelined 4-bit ripple carry adder is constructed using the one-bit full adder [designed in part (a)], using the method shown in In Fig. 6b,

The pipeline is clocked at 1GHz.

The timing parameters of latch is as follows-

Set up = 25ps, CLK-to-Q delay = 50 ps ,

D to Q delay = 40 ps , hold time = 30 ps,

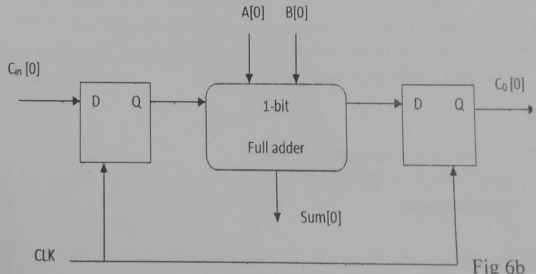


Fig 6b

Estimate the maximum delay permissible for the 1-bit full adder.

- c) Find the W/L ratios of all MOSFETs in the 1-bit full adder circuit designed in part (a) such

that the 1-bit full adder can be included in the pipeline as mentioned in part (b).

Assume that individual output nodes (of dual rail domino and DCVSL block) has a total load capacitance of 0.5fF.

Clearly explain the analysis done for sizing the MOSFETs.

[20]